AmarVote

Secure Cryptographic Voting Platform

Guardian Guide

Your responsibilities as a Guardian:

- Account Registration & Login
- Phase 1: Key Pair Generation
- Phase 2: Backup Share Generation
- Partial & Compensated Decryption
- Combining Decryption Shares

This guide contains critical security instructions.
Read every warning box carefully.

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Your Role as a Guardian

You have been assigned as a **Guardian** for an AmarVote election. This is a position of significant trust and responsibility.

What does a Guardian do?

In AmarVote, no single person can decrypt the election results alone. Instead, the decryption key is mathematically distributed among all guardians using a technique called *threshold cryptography*. To decrypt the tally and reveal the results, a minimum number of guardians (the *quorum*) must cooperate.

Why is this important?

This means:

- Even the election administrator cannot manipulate or see individual votes.
- Even if one guardian is malicious, they cannot singlehandedly decrypt the tally.
- Even if some guardians are unavailable, the election can still be tallied (if enough guardians cooperate to meet the quorum).

Important

Your two most critical responsibilities:

1. **Securely store** your encrypted private key file (downloaded in Phase 1).
2. **Remember or safely store** your Guardian Password.

If you lose either of these, you may prevent the election results from ever being published.

Chapter 1

Account Setup: Registration & Login

Before performing any guardian duties, you need a registered AmarVote account.

1.1 Prerequisites

Your email address must be on the **Authorised User List**. The election administrator or platform Owner will have added you. If you are unsure, contact the election administrator.

1.2 Registering Your Account

1. Open your browser and navigate to the AmarVote URL.
2. Click **Register** or go to `/register`.
3. Enter your authorised email address in the `Email Address` field.
4. Click **Send Verification Code**.
5. Check your email inbox for a 6-digit One-Time Password (OTP) from AmarVote. Check your spam folder if you don't see it within 2 minutes.
6. Enter the OTP in the `Verification Code` field.
7. Click **Verify Email**.
8. Create a strong password (12+ characters, mixed case, numbers, symbols).
9. Confirm the password in the `Confirm Password` field.
10. Click **Create Account**.

Critical

As a Guardian, your account credentials protect the election's cryptographic infrastructure. Use a strong, unique password and store it in a password manager. Do not reuse passwords from other services.

1.3 Enabling Two-Factor Authentication

2FA is strongly recommended for all guardian accounts.

1. After logging in, click your avatar and select **Profile Settings**.
2. Under **Security**, click **Enable Two-Factor Authentication**.
3. Scan the QR code with Google Authenticator or any TOTP app.
4. Enter the current 6-digit code from the app and click **Activate**.
5. **Save the displayed recovery codes** in a secure location.

1.4 Logging In

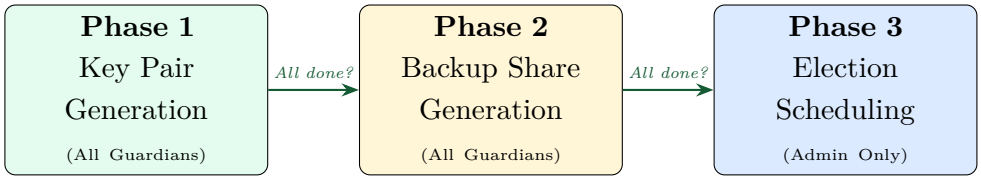
1. Navigate to the AmarVote URL and click **Login**.
2. Enter your `Email Address` and `Password`.
3. If 2FA is enabled: enter the 6-digit TOTP code from your authenticator app.
4. Click **Sign In**.

Chapter 2

Understanding the Key Ceremony

Before diving into the steps, it is worth understanding *what* you are doing and *why*. This chapter gives a conceptual overview.

2.1 The Three-Phase Ceremony



Cryptographic Concept

Asymmetric cryptography basics:

- A **key pair** consists of a **public key** and a **private key**.
- Anything encrypted with a public key can *only* be decrypted with the matching private key.
- In AmarVote, ballots are encrypted with a combined election public key. Decryption requires guardian private keys (above the quorum threshold).

2.2 What Happens to Your Key

- **Your public key** is stored in the AmarVote database and used in ballot encryption.
- **Your private key** is *never* stored on the server. It is encrypted with your password using AES-256 and stored only on your device.
- **Backup shares** of your private key are split into $(n-1)$ fragments (where n is the total number of guardians) and each fragment is encrypted with a different guardian's public key. These encrypted fragments are stored in the database.

Note

Why backup shares matter: If you cannot participate in decryption, the other guardians can use your backup shares (which only they can decrypt) to reconstruct your key contribution. This is how the system tolerates absent guardians while

maintaining the quorum requirement.

Chapter 3

Phase 1: Key Pair Generation

Phase 1: Key Pair Generation

When: Immediately after election creation.
Who: All guardians (each independently).
Blocking: Phase 2 cannot begin until *all* guardians complete Phase 1.
Output: One public key (uploaded to server) + one encrypted private key file (downloaded to your device).

3.1 Finding Your Election

1. Log in to AmarVote.
2. You may see a notification in the Notifications area indicating you have a guardian assignment. Click it to go directly to the election, *or*:
3. Navigate to **All Elections** in the sidebar.
4. Find the election for which you are a guardian and click on it to open the election page.

3.2 Accessing the Guardian Tab

1. On the election page, look for the tab bar near the top.
2. Click the **Guardian** tab.
3. You will see the Key Ceremony status panel. Phase 1 should be marked as **Active**.

3.3 Generating Your Key Pair

1. In the Phase 1 panel, click **Generate Key Pair**.
2. The system will cryptographically generate your unique asymmetric key pair for this election. This happens entirely in your browser — the private key never leaves your device unencrypted.
3. A password prompt will appear with two options:

3.3.1 Setting Your Guardian Password

Option	What It Does
Generate Secure Password	The system generates a cryptographically random, high-entropy password. Recommended.
Enter Your Own Password	You type a password of your choice. Must be strong (see requirements below).

Critical

This password is irreplaceable. There is no "forgot password" for the guardian password. If you forget it, your private key file is permanently inaccessible, and you will be unable to participate in decryption.

Tip

Use the **Generate Secure Password** button. Then immediately copy the generated password and save it in your password manager with a label like `AmarVote Guardian Password - [Election Name]`.

3.4 Submitting and Downloading

1. Once you have set and recorded your password, click **Submit Key Pair & Download**.
2. Two things happen simultaneously:
   (a) Your **public key** is transmitted to the AmarVote server and stored in the database.
   (b) An **encrypted private key file** (e.g., `guardian_key_[yourname]_[election_id].enc`) is automatically downloaded to your browser's default downloads folder.
3. A green success message will appear: "Phase 1 Complete".

3.5 Immediately After Downloading

1. Move the downloaded `.enc` file to a **secure, backed-up location**:
   - An encrypted USB drive (stored securely offline)
   - An encrypted cloud storage folder (e.g., encrypted with VeraCrypt before upload)
   - Your system's encrypted home directory (ensure full-disk encryption is enabled)

2. Do **not** leave it in your Downloads folder indefinitely.
3. Verify your password is saved in your password manager.

Important

Do not share the private key file or your guardian password with anyone — including the election administrator. Sharing these would compromise the security guarantee of the election.

Chapter 4

Phase 2: Backup Share Generation

Phase 2: Backup Share Generation

When: After all guardians complete Phase 1.
Who: All guardians (each independently).
Blocking: Phase 3 (election scheduling) cannot begin until all guardians complete Phase 2.
Output: Encrypted backup shares stored in the server database.

4.1 What Backup Shares Are

Cryptographic Concept

Shamir's Secret Sharing (simplified): Your private key is mathematically split into $n - 1$ fragments (where n is the total number of guardians). Each fragment is encrypted using the public key of a different guardian. Only that guardian can decrypt their fragment. If quorum-many guardians each provide their fragment, the original private key can be reconstructed — but no single fragment reveals anything on its own.

Example with 5 guardians, quorum = 3:

- Guardian G1 generates 4 backup fragments (for G2, G3, G4, G5).
- The fragment for G2 is encrypted with G2's public key — only G2 can open it.
- If G1 is missing during decryption, G2, G3, and G4 can each provide their fragment of G1's key. With 3 fragments (meeting quorum), G1's key is reconstructed.

4.2 Generating Your Backup Shares

1. Navigate to the election page and open the **Guardian** tab.
2. Phase 2 should now show as **Active**.
3. Click **Upload Encrypted Key File** and select the `.enc` file you downloaded in Phase 1.

4. Enter your `Guardian Password` in the provided field.
5. Click **Generate Backup Shares**.
6. The browser will:
   (a) Decrypt your private key in memory using your password (via AES-256).
   (b) Generate $n - 1$ backup fragments from your private key.
   (c) Encrypt each fragment with the corresponding guardian's public key.
   (d) Transmit all encrypted fragments to the server.
   (e) Erase the decrypted private key from memory.
7. A success message will appear: "Phase 2 Complete".

Note

The decryption of your private key happens *entirely within your browser*, in memory, for only the duration of this process. Your password and private key are **never sent to the server**. Only the encrypted backup shares (which are meaningless without the recipient's private key) are transmitted.

Important

If you close the browser or lose your internet connection during backup share generation, the process may fail. Return to the Guardian tab and repeat the step — it is safe to retry.

4.3 After Completing Phase 2

After you submit your backup shares:

- You do not need to take any further action until the election closes.
- You may optionally vote in the election if you are also on the voter list.
- Keep your private key file and password safe — you will need them again for decryption.

Chapter 5

Waiting Period: During the Election

After Phase 2, the election administrator will schedule the election and voters will cast their ballots. During this period, your guardian responsibilities are paused.

5.1 What to Do During This Period

- **Keep your credentials safe**: Your private key file and guardian password must remain accessible to you after the election closes.
- **Watch for notifications**: AmarVote will notify you when the election closes and the tally phase begins.
- **Participate as a voter**: If you are on the voter list, you may cast your own ballot just like any other voter.

> **Important**
>
> If you change devices, transfer your encrypted private key file to the new device *before* the decryption phase begins. Ensure your password manager is synced.

Chapter 6

Post-Election: Partial Decryption

After the election closes and the tally is computed (initiated by the admin), each guardian must perform their partial decryption. This is the step where your private key is used to "unlock" your share of the tally.

6.1 When to Begin

- You will receive a notification that the tally is complete and decryption can begin.
- Navigate to the election page and open the **Guardian** tab.
- You will see the **Partial Decryption** panel is now active.

6.2 Performing Partial Decryption

1. Click **Upload Encrypted Key File**.
2. Select the `.enc` file from your secure storage location.
3. Enter your `Guardian Password` in the provided field.
4. Click **Submit Credentials & Decrypt**.

6.2.1 What Happens During Decryption

The browser performs the following operations:

1. **Credential Verification**: Your password decrypts your private key in memory. If the password is incorrect, an error is shown immediately.
2. **Partial Decryption**: Your private key is applied to each chunk of the encrypted tally. This produces a *partial decryption* for your guardian share — a mathematical result that, when combined with others, will reveal the total.
3. **Compensated Decryption**: For each *other* guardian's backup share of *your* key (stored in the database), your private key decrypts those shares and computes *compensated decryption values*. These substitutes allow the system to account for guardians who cannot participate, without revealing their private keys.

4. **Submission**: All partial and compensated decryption data is submitted to the server.
5. **Memory Cleanup**: Your decrypted private key is erased from browser memory.

Important

Do not close the browser tab during this process. Partial decryption involves significant computation and may take several minutes depending on the number of ballot chunks (each chunk contains approximately 200 ballots).

Note

Progress tracking: A progress bar will display the current chunk being processed vs. the total number of chunks. For a 2000-ballot election with 200-ballot chunks, there would be 10 chunks to process.

6.3 After Partial Decryption is Complete

1. A success message will confirm your partial and compensated decryptions have been submitted.
2. Your guardian contribution is complete.
3. The election page will show updated progress: how many of the required quorum have now submitted their decryptions.
4. Once the quorum is met, either you or the admin can click **Combine Decryption Shares**.

Chapter 7

Combining Decryption Shares

When at least *quorum* guardians have completed their partial decryption, the final results can be produced.

7.1 Who Can Combine

Both the election administrator and any guardian can initiate the combination step.

7.2 Steps to Combine

1. Navigate to the election page's **Tally** or **Guardian** tab.
2. Check the decryption status table — confirm that the required number of guardians (at least quorum) show as **Complete**.
3. Click **Combine Decryption Shares**.
4. The server will:
   (a) Retrieve all submitted partial decryptions.
   (b) For absent guardians (below quorum threshold), substitute with compensated decryptions.
   (c) Mathematically combine all shares using the ElectionGuard protocol.
   (d) Produce the final plaintext vote totals.
   (e) Publish the results with full cryptographic proofs.
5. The results page will load automatically when the combination is complete.

Important

Verify before combining. Once you click **Combine Decryption Shares**, the results are final and published. Make sure the correct number of guardians have completed decryption before proceeding.

Chapter 8

Verifying the Results

As a guardian, you have a special responsibility to review the published results and confirm that the cryptographic proofs are valid.

8.1 Reviewing Published Results

1. After combination, navigate to the **Results** tab of the election.
2. Review the vote totals, percentage breakdown, and total ballot count.
3. Compare these results with your own expectations (if you monitored the election).

8.2 Downloading the Audit Package

1. Navigate to the **Verification** tab.
2. Click **Download Election Data**.
3. The package contains all cryptographic artefacts except private keys.
4. Optionally, verify the data using ElectionGuard's open-source verifier tools.

Tip

As a guardian who understands the cryptographic process, you are well-positioned to independently verify the election using ElectionGuard's reference verifier. This is the gold standard of election auditability.

Appendix A

Guardian Security Checklist

Complete this checklist before, during, and after your guardian duties:

Before Phase 1:

☐ Registered AmarVote account with authorised email

☐ Enabled 2FA with Google Authenticator

☐ Set up password manager to store guardian password

☐ Identified a secure storage location for the private key file

After Phase 1:

☐ Guardian password saved in password manager

☐ Private key file (`.enc`) downloaded

☐ Private key file moved to secure storage (not Downloads folder)

☐ Private key file backed up in at least one additional secure location

After Phase 2:

☐ Backup shares successfully generated and submitted

☐ Confirmed Phase 2 status shows "Complete" for your guardian entry

☐ Private key file and password still securely stored (do not delete them!)

After Election Closes (Decryption Phase):

☐ Located private key file in secure storage

☐ Retrieved guardian password from password manager

☐ Completed partial and compensated decryption

☐ Confirmed decryption submission success message

☐ If quorum is met: clicked Combine or confirmed admin has done so

Appendix B

Frequently Asked Questions for Guardians

What if I forget my Guardian Password?

Unfortunately, there is no password recovery mechanism for the guardian private key. Your password is not stored anywhere on the server. If you forget your password, your encrypted private key file is permanently inaccessible.

If your absence would cause the available guardians to fall below the quorum threshold, contact the election administrator immediately. They can assess whether compensated decryption from other guardians can substitute for your share.

What if I lose my private key file?

If you lose the `.enc` file but still have your guardian password (or vice versa), neither is usable without the other. The situation is the same as forgetting the password — contact the administrator immediately.

Can I participate in decryption from a different device?

Yes. Transfer your encrypted `.enc` file to the new device (e.g., via an encrypted USB drive or secure file transfer). Ensure your guardian password is accessible (e.g., via a synced password manager). Then proceed to the Guardian tab on the new device as normal.

Can I see the election results before they are published?

No. During your partial decryption, you produce a *mathematical share* of the result, not the result itself. The actual totals are only revealed when all shares are combined. This prevents any single guardian from learning the results before official publication.

What if the tally takes a very long time?

The tally and decryption processes are computationally intensive. For large elections with thousands of ballots, the process may take 10–30 minutes or more. The progress bar on the Guardian tab provides real-time updates. Do not interrupt the process.